Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/13/2004
PI Name:	Louge, Michel Y. Ph.D.		
Project Title:	Studies of Gas - Particle Interactions in a Microgravity Flow Cell		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853-7501	**Congressional District:**	22
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	02/01/1998	**End Date:**	12/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	10	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jenkins, James (Cornell University) Reeves, Anthony (Theoretical & Applied Mechanics)		
Grant/Contract No.:	NAG3-2705		
Performance Goal No.:	2.5		
Performance Goal Text:	Advance the state of knowledge sufficiently to enable dust control technologies and bulk materials handling for extraterrestrial habitats and/or In Situ Resource Utilization (ISRU).		

Task Description:

This flight project is also known as "SiGMA" (Solids Interacting with a Gas in a Microgravity Apparatus).
This research is made possible by the development of a unique "Couette" apparatus producing shearing flows of gas and agitated solids in the absence of gravitational accelerations. The facility, if incorporated as originally planned in the Granular Flow Module of the International Space Station, would permit gas-particle interactions to be observed over a range of conditions where suspensions are steady and fully-developed.

Unlike Earth-bound flows where the gas velocity must be set to a value large enough to defeat the weight of particles, the duration and quality of microgravity on the Space Station would permit us to achieve suspensions where the agitation of the particles and the gas flow can be controlled independently by adjusting the gas pressure gradient along the flow and the relative motion of the boundaries.

We envision two series of experiments in the new apparatus. In the first series, which we call "viscous dissipation experiments," we characterize the viscous dissipation of the energy of the particle velocity fluctuations, when there is no relative mean velocity between gas and solids. To do so, we can reduce the boundary speed in successive tests until the inertia of the solid particles becomes small enough for the particle motion to be affected by viscous forces in the gas. By evacuating the apparatus partially, we can also investigate the role of the molecular mean free path in reducing the particle agitation.

In a second series of tests, which we call "viscous drag experiments," we can impose a gas pressure gradient on the flow channel. The gradient induces a relative velocity between the two phases, while the shearing will set the solids agitation independently. These Viscous Drag Experiments are unique in exploring a regime where particle velocity fluctuations are determined by a mechanism other than interactions with the gas. In this regime, we can measure the dependence of the drag coefficient on the solid volume fraction and agitation of the solid particles. Partially evacuation would also allow us to test the effects of particle Reynolds number on the drag coefficient.

To conduct these tests, NASA has designed an axisymmetric "Couette" apparatus consisting of two counter-rotating boundaries to which half-cylindrical bump are affixed. Observations can be made through the top window using a digital video camera. The velocity of the flow spheres can be inferred using new computer vision techniques.

To prepare for flights on the International Space Station, we designed and built a prototype apparatus that we successfully tested on the KC-135 microgravity aircraft. The lessons learned from the operations of the prototype have informed the design of the space experiment.

This project’s objective is to observe details of the interactions between a gas and agitated solid particles, and to generate experimental data to validate numerical simulations and theories for gas-solid flows.

Flows of gases loaded with granular materials occur in nature and in industrial processes. On Earth, applications are found in the chemical, mining and pharmaceutical industries. The long-term human or robotic exploration of the Moon and Mars requires the development of “In-Situ Resource Utilization” (ISRU) for propellant production, habitat, infrastructure, and extraction of water and breathable gas. These new technologies all require a deeper understanding of interactions among grains and between grains and gases. On the particle scale, viscous hydrodynamic interactions and grain collisions are a crucial attribute of the physics.

By validating recent theories and numerical simulations for the interactions between a gas and agitated solid particles, this work will improve the feasibility of planetary missions where fuel and water must be extracted from mining the soil. It will also benefit sectors as diverse as the pharmaceutical, oil, mining, and chemical engineering industries.

In-Situ Resource Utilization will involve moving granular powders or agglomerates from mine to plant or within a plant by pneumatic transport in closed pipes. Because the drag force on agitated aggregates of particles is not well characterized and the mechanisms by which energy is lost are poorly understood, it is impossible to design processing equipment to eliminate jamming in dense flows with absolute certainty.

Upon scale-up to realistic plant size, pneumatic transport is known to develop clustering instabilities that can lower the average drag on the particulate suspension by an order or magnitude from that of a homogeneous suspension of the same mean concentration (Grace, et al., 1997; Agrawal, et al. 2001). The clusters can also merge into “ropes” that concentrate the particles within narrow bands aligned with the flow. Such ropes make it very difficult to predict the behavior of passive gas-solid separators such as cyclones (Bricout and Louge, 2004a), which are used downstream of a transport line or reactor to extract clean gaseous products from the suspension. The interactions of suspended solids with boundaries also affect the overall flow behavior, such as the descent of clusters at the wall of a circulating fluidized bed (Grace, et al., 1997), and can cause extensive damage to surfaces by wear. All such phenomena depend on gravity, particle size, particle shape, electrostatics, environmental conditions and unit size in ways that are not yet predictable without extensive tests on realistic pilot plants.

The necessary research involves the study of two important aspects of gas-solid flows over a range of flow conditions: the characterization of the drag exerted by the gas on agitated particles, and the determination of the amount of energy transferred between the gas and the particles. This will permit the rational design of pneumatic-transport lines and fluidized beds and permit a description of Aeolian transport of sand and the settling of dust on Mars.

In this project, NASA has designed experiments for the GFM in the International Space Station (ISS) to measure gas drag and the viscous dissipation of particle agitation energy over a range of carefully controlled flow conditions.

We have planned two series of experiments in the flight apparatus. In the first series, which we call "viscous dissipation experiments," we can characterize the viscous dissipation of the energy of the particle velocity fluctuations, when there is no relative mean velocity between gas and solids. To do so, we reduce the boundary speed in successive tests until the inertia of the solid particles becomes small enough for the particle motion to be affected by viscous forces in the gas.

In a second series of tests, which we call "viscous drag experiments, "we can impose a gas pressure gradient on the flow channel. The gradient induces a relative velocity between the two phases, while the shearing sets the solids agitation independently. These Viscous Drag Experiments are unique in exploring a regime where particle velocity fluctuations are determined by a mechanism other than interactions with the gas. In this regime, we can measure the dependence of the drag coefficient on the solid volume fraction and agitation of the solid particles.

To conduct these tests, NASA has developed an axisymmetric "Couette" apparatus consisting of two counter-rotating boundaries to which half-cylindrical bump are affixed. Observations can be made through the top window using a digital video camera. The velocity of the flow spheres can be inferred using new computer vision techniques. The apparatus can

	produce shearing flows of gas and agitated solids in the absence of gravitational accelerations. The facility, if incorporated in the Granular Flow Module of the International Space Station, would permit gas-particle interactions to be observed over a range of conditions where suspensions are steady and fully-developed. Unlike Earth-bound flows where the gas velocity must be set to a value large enough to defeat the weight of particles, the duration and quality of microgravity on the Space Station can achieve suspensions where the agitation of the particles and the gas flow can be controlled independently by adjusting the gas pressure gradient along the flow and the relative motion of the boundaries. To prepare for flights on the International Space Station, we have designed and built a prototype apparatus that we tested successfully on the KC-135 microgravity aircraft. The lessons learned from the operations of the prototype have informed the design of the space experiment that was carried out by Zin Technologies under contract from the NASA Glenn Research Center. Numerical simulations using the Lattice-Boltzmann (LB) technique (Ladd and Verberg, 2001; Xu, et al., 2004) will soon be able to capture our Viscous Dissipation and Viscous Drag experiments. Those experiments then serve as a unique, well-controlled tool to validate the simulations. Once validated, the LB simulations can be used to establish constitutive relations for gas-solid suspension involving realistic particle shapes and size distribution. In turn, the constitutive relations can be used to create models of larger systems for practical applications, such as those necessary for the exploration of the Moon and Mars.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The International Fine Particle Research Institute (IFPRI), a group of some of the leading companies involved with powder flows, has provided support for, and has monitored progress of this project. IFPRI currently includes 28 member companies, some of which are well-known in Chemical or Pharmaceutical Engineering: Agfa-Gevaert, Akzo Nobel, Alcoa, Dow Chemical, du Pont, ExxonMobil, GlaxoSmithKline, Merck, Nestlé, Pfizer, Procter, Gamble, etc. The commitment of these companies is a clear endorsement of the feasibility and importance of this project.
Task Progress:	We now list papers written or published in 2004 and talks presented. Papers Xu H, Reeves AP, Louge MY. “Measurement Errors in the Mean and Fluctuation Velocities of Spherical Grains from a Computer Analysis of Digital Images.” Review of Scientific Instruments, 75, 811-819 (2004). Louge M. “Role of Couple Stresses in Shallow Granular Flows down a Bumpy Incline.” 5th International Conference on Multiphase Flow, ICMF’04, Yokohama, Japan, May 30-June 4, 2004, Paper No.K13. Louge M, Xu H. “Collisional Granular Flows with and without Gas Interactions in Microgravity,” Mini-Symposium on “Microgravity Flow Phenomena,” 21st International Congress of Theoretical and Applied Mechanics, August 15-21, Warsaw, Poland, in press. Xu H, Verberg R, Koch DL, Louge M. “Dense bounded shear flows of agitated solid spheres in a gas at intermediate Stokes and finite Reynolds numbers, submitted to the J. Fluid Mech., May 25, 2004. Allen Wilkinson R, Robert P. Behringer, James T. Jenkins, and Michel Y. Louge: “Granular Materials and the Risks They Pose for Success on the Moon and Mars,” paper prepared for STAIF 2005. Louge M. “Scientific issues in the flow of gases with dispersed solids” Multiphase Science and Technology, Vol. 15, Nos. 1-3, pp. 1-5, 2003. Talks Louge M. “On dense granular flows down bumpy inclines,” invited talk, Purdue University, February 5, 2004. Louge M. “Role of Couple Stresses in Shallow Granular Flows down a Bumpy Incline,” 5th International Conference on Multiphase Flow, ICMF’04, Yokohama, Japan, keynote speech, Wednesday, June 2, 2004. Louge M. GDR Midi, Carry-le-Rouet, France “Le rôle du moment angulaire dans les écoulements gravitaires,” invited talk, June 8, 2004. Louge M. Conference on Strategic Research to Enable NASA’s Exploration Missions, invited talk on SiGMA and mugSEG, June 23, 2004. Louge M. “Collisional Granular Flows with and without Gas Interactions in Microgravity,” Mini-Symposium Invited lecture at ICTAM 2004, Warsaw, Poland in the mini-symposium on “Microgravity Flow Phenomena”, 21st International Congress of Theoretical and Applied Mechanics, August 15-21, Warsaw, Poland, August 16, 2004.
Bibliography Type:	Description: (Last Updated: 10/29/2019)
Articles in Peer-reviewed Journals	Louge M. "Scientific issues in the flow of gases with dispersed solids." Multiphase Science and Technology 2003; 15(1-3): 1-5. , Oct-2003
Articles in Peer-reviewed Journals	Xu H, Reeves AP, Louge MY. "Measurement Errors in the Mean and Fluctuation Velocities of Spherical Grains from a Computer Analysis of Digital Images." Review of Scientific Instruments. 2004; 75, 811-819. https://doi.org/10.1063/1.1666989 , Mar-2004
NASA Technical Documents	M. Louge "SiGMA and mugSEG" : Conference on Strategic Research to Enable NASA’s Exploration Missions, invited talk on SiGMA and mugSEG, June 23, 2004. Jun-2004

Papers from Meeting Proceedings	Louge M. "Role of Couple Stresses in Shallow Granular Flows down a Bumpy Incline." 5th International Conference on Multiphase Flow, ICMF'04, Yokohama, Japan, May 30-June 4, 2004, Paper No.K13. 5th International Conference on Multiphase Flow, ICMF'04, Yokohama, Japan, May 30-June 4, 2004, Paper No.K13. , Jun-2004
Papers from Meeting Proceedings	Louge M, Xu H. "Collisional Granular Flows with and without Gas Interactions in Microgravity." Mini-Symposium on "Microgravity Flow Phenomena," 21st International Congress of Theoretical and Applied Mechanics, August 15-21, Warsaw, Poland. The Proceedings, entitled "Mechanics of the 21st Century" and edited by W. Gutkowski and T.A. Kowalewski, have been published by Springer, Dordrecht, The Netherlands, 2005. Mini-Symposium on "Microgravity Flow Phenomena," 21st International Congress of Theoretical and Applied Mechanics, August 15-21, Warsaw, Poland. The Proceedings, entitled "Mechanics of the 21st Century" and edited by W. Gutkowski and T.A. Kowalewski, have been published by Springer, Dordrecht, The Netherlands, 2005. , Aug-2004
Papers from Meeting Proceedings	Allen Wilkinson R, Behringer RP, Jenkins JT, Louge MY. "Granular Materials and the Risks They Pose for Success on the Moon and Mars." AIP Conference Proceedings. 2005; 746, 1216. https://doi.org/10.1063/1.1867248 AIP Conference Proceedings. 2005; 746, 1216. https://doi.org/10.1063/1.1867248 , Mar-2005
Presentation	M. Louge " "On dense granular flows down bumpy inclines,"" invited talk, Purdue University, February 5, 2004. Feb-2004
Presentation	M. Louge ": "Role of Couple Stresses in Shallow Granular Flows down a Bumpy Incline," " 5th International Conference on Multiphase Flow, ICMF'04, Yokohama, Japan, keynote speech, Wednesday, June 2, 2004 Jun-2004
Presentation	M. Louge ""Le rôle du moment angulaire dans les écoulements gravitaires,"" GDR Midi, Carry-le-Rouet, France, invited talk, June 8, 2004. Jun-2004
Presentation	M. Louge " "Collisional Granular Flows with and without Gas Interactions in Microgravity,"" Mini-Symposium Invited lecture at ICTAM 2004, Warsaw, Poland in the mini-symposium on "Microgravity Flow Phenomena", 21st International Congress of Theoretical and Applied Mechanics, August 15-21, Warsaw, Poland, August 16, 2004. Aug-2004